

MTU - CORK INSTITUTE OF TECHNOLOGY
INSTITIÚID TEICNEOLAÍOCHTA CHORCAÍ

Semester Project 2021/22

Module Title: Times Series & PCA

Module Code: STAT9005

School: Mathematical Science and Informatics

Programme Title: MSc in Science in Data Science and Analytics

Internal Examiner(s): Francisco Hernandez

Instructions:

This assessment is worth **40%** of your final grade in STAT9005.

(**30%**) Report on case studies (**20% for Time Series** and 10% for PCA).

(**10%**) Recorded Presentation.

Deadline: Submit via Canvas the report in PDF format, the RStudio script and a ten minutes presentation of the work, no later than **Sunday 22nd of May 2022** at 22 hours. The final file submitted should include the report, code and presentation for both parts (Time Series and Dimensionality Reduction).

You should also include a **declaration** as part of your submission that all work contained in the submission was your own except for others work which is clearly referenced.

TIME SERIES ANALYSIS

The dataset of the *“household_power_consumption.txt”* contains 2,075,259 measurements gathered in a house located in Sceaux (7km of Paris, France) between December **2006** and November **2010 (47 months)**. The measurements of the household are taken with a **one-minute sampling rate** over the almost 4 years period. There are different measurements from different electrical quantities. The columns are:

1.date: Date in format dd/mm/yyyy

2.time: time in format hh:mm:ss

3.global_active_power: household global minute-averaged active power (in kilowatt)

4.global_reactive_power: household global minute-averaged reactive power (in kilowatt)

5.voltage: minute-averaged voltage (in volt)

6.global_intensity: household global minute-averaged current intensity (in ampere)

7.sub_metering_1: energy sub-metering No. 1 (in watt-hour of active energy). It corresponds to the kitchen, containing mainly a dishwasher, an oven and a microwave (hot plates are not electric but gas powered).

8.sub_metering_2: energy sub-metering No. 2 (in watt-hour of active energy). It corresponds to the laundry room, containing a washing-machine, a tumble-drier, a refrigerator and a light.

9.sub_metering_3: energy sub-metering No. 3 (in watt-hour of active energy). It corresponds to an electric water-heater and an air-conditioner.

The analysis of this project will be focused on the **monthly averaged global_active_power** variable. **Analyse the Time Series** of the **monthly average global active power consumption** in *RStudio*. Use the libraries available in R which you consider are appropriate for this study.

More information about the dataset can be found in the following site:

[UCI Machine Learning Repository: Individual household electric power consumption Data Set](#)

Data cleaning and manipulation

Missing values of the variable *global_active_power* need to be estimated by **interpolation** of previous and posterior observations.

Using the *global_active_power* energy consumed **every minute** (in watt hour) in the household by electrical equipment, create a new variable for the global active power representing the **monthly average consumption**.

Preliminar analysis

- Carry out a **preliminary descriptive** analysis on the data. Use summary statistics and plots to describe the dataset.
- **Decompose** the dataset and analyse the different components of the series. State what type of model – additive or multiplicative – is more appropriate to describe this series. Help your conclusions by using appropriate plots and summaries.

- Analyse which of the **components** (if any) has a strongest impact on the time series. Is this component conditioning the analysis in any way? Explain how this issue could be addressed if required.
- Indicate if **periodicity** could be a problem in the time series.

Time series modelling

- Implement the **Classical Method** (*SES, Holt's, Holt-Winter's*) that you consider is more appropriate for describing this data. Explain why do you think this is the best option.

Implement your chosen classical model and indicate the best model options for the trend and seasonality components (**additive** and **multiplicative**). Explain which of the options **fits better** the time series. Support your conclusions.

Is this conclusion consistent with what you expected in the preliminary analysis?

- Indicate whether this time series is **stationary**. Use the formal tests and graphical tools you consider appropriate to support your conclusion.

In case it is **not stationary**, manipulate the dataset to obtain an appropriate series which meets the stationarity conditions for further analysis using **ARIMA** models. Explain and justify all the process in the transformation.

Use the **correlograms** for analysing the transformed time series. Based on the correlograms, what type of ARIMA-SARIMA model do you expect to best describe this dataset. Explain your answer.

Chose **three** models you think are good **candidates** for this dataset and **analyse the residuals**. Use plots and formal tests to compare the models.

Find the model proposed by R using "**auto.arima**" command and discuss whether you think this is a good option for this dataset. Do you think that any of your models would be a better option?

What **model** do you consider is **better** for this dataset, the Classical or the ARIMA model?

Forecast

Make a **12-month forecast** for the next periods in both cases, Classical and ARIMA models.

- In the **Classical** model, carry out the **prediction** for multiplicative and additive scenarios and compare the results.
- For the **ARIMA** models, carry out the **prediction** using the *auto.arima* models recommended by R and the model you chose during your analysis. Compare the results.
- Make a **comparison** of the **results** in the predictions using the Classical and the ARIMA models.
- Check the **power of prediction** of your models by forecasting the last periods of your time series and **compare** them with the actual **observed values** for such period you have predicted.